PEFS-2: Prime-Evolution Feedback System

This document formalizes the **Prime-Evolution Feedback System** (PEFS) and its quantum-control-ready variants, incorporating:

- Prime-indexed channel decomposition and truncation (“prime bandwidth”)
- A **norm / Lipschitz ledger** for global stability (discrete) and dissipativity-ledger (continuous)
- Physically correct quantum embeddings (CPTP/open-system and unitary/closed-system)
- Noncommutation monitoring via commutator budgets

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0. Notation and standing assumptions

- H: complex separable Hilbert space.
- $\mathcal{B}(H)$: bounded linear operators on H.
- $\|\cdot\|$: operator norm on $\mathcal{B}(H)$ and induced norm on H.
- $\mathbb{P}$: set of primes.
- **Prime bandwidth** $P \in \mathbb{N}$: truncation parameter; sums are over $p \le P$.

Nonlinearity

- $T : H \to H$ is globally Lipschitz with constant L_T: $$ |T(x)-T(y)|\le L_T\,|x-y|\quad\forall x,y\in H. $$

Real multiplicity injection

- $\Lambda_m : \mathbb{T} \to \mathbb{R}$ (discrete $t \in \mathbb{Z}_{\ge 0}$ or continuous $\tau \in \mathbb{R}_{\ge 0}$).
- When computed from complex prime contributions $\Lambda_m^{(p)}$, enforce reality by $$ \Lambda_m(t)=\Re\Big(\sum_{p\le P}\Lambda_m^{(p)}(t)\Big), $$ using conjugate pairing or a specified summation/regularization rule.

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1. Discrete-time PEFS-2 (state-space dynamics)

1.1 Prime-banded linear channel

Choose operators $U_p(t) \in \mathcal{B}(H)$ and scalar weights $w_p(t) \in \mathbb{C}$ (or $\mathbb{R}$) and define

$$ \Xi_P(t):=\sum_{p\le P} w_p(t)\,U_p(t)\in\mathcal{B}(H). $$

Absolute convergence is automatic for finite P; for $P = \infty$ require $\sum_p |w_p(t)|\,\|U_p(t)\| < \infty$ uniformly in t.

1.2 Update rule

$$ X_{t+1} = \Xi_P(t)X_t + \Lambda_m(t)\,T(X_t),\qquad X_0\in H. $$

1.3 Ledger condition (uniform contraction)

Define the **one-step Lipschitz constant**

$$ q:=\sup_t\Big(|\Xi_P(t)| + |\Lambda_m(t)|\,L_T\Big). $$

Ledger assumption: $q < 1$.

1.4 Consequences (well-posedness and stability)

Under $q < 1$:

1. Each step map $\Phi_t(x) = \Xi_P(t)x + \Lambda_m(t)T(x)$ is a contraction with constant $\le q$.
2. Uniqueness and exponential forgetting: $$ |X_t-Y_t|\le q^t\,|X_0-Y_0|\quad\forall t. $$
3. Robustness to bounded disturbances b_t: if $X_{t+1} = \Phi_t(X_t) + b_t$ with $\sup_t \|b_t\| < \infty$, then trajectories remain uniformly bounded with a bound scaling like $\sup \|b_t\|/(1-q)$ (plus a transient).

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2. Continuous-time PEFS-2 (generator-correct form)

2.1 Prime-decomposed generator

Instead of differentiating Ξ directly, specify a time-dependent generator

$$ A_P(\tau):=\sum_{p\le P} a_p(\tau)\,G_p(\tau), $$

where $G_p(\tau) \in \mathcal{B}(H)$ (bounded case) and $a_p(\tau) \in \mathbb{R}$ (or complex with an explicit reality policy).

2.2 ODE / mild evolution

$$ \dot X(\tau)=A_P(\tau)X(\tau)+\Lambda_m(\tau)\,T(X(\tau)),\qquad X(0)=X_0. $$

2.3 Well-posedness (bounded case)

If $\sup_{\tau\ge 0} \|A_P(\tau)\| < \infty$ and T is globally Lipschitz, then the system has a unique global solution (Picard–Lindelöf in Banach spaces).

2.4 Stability ledger (dissipativity)

If there exists $\alpha > 0$ such that

$$ \Re\langle A_P(\tau)x,x\rangle\le -\alpha|x|^2\quad\forall x\in H,\ \forall\tau, $$

and

$$ \sup_{\tau\ge0}|\Lambda_m(\tau)|\,L_T < \alpha, $$

then differences contract exponentially.

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3. Truncation and “prime bandwidth” as a model parameter

3.1 Tail bound (operator norm)

For an absolutely convergent infinite series $\Xi(t) = \sum_p w_p(t)U_p(t)$, define Ξ_P as the $p \le P$ truncation. Then

$$ |\Xi(t)-\Xi_P(t)|\le \sum_{p>P} |w_p(t)|\,|U_p(t)|, $$

uniformly in t when the tail bound is uniform.

3.2 Operational interpretation

- P is the **prime bandwidth** (compute budget).
- Tail bounds supply formal **approximation guarantees**.

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4. Spectral alignment and sector structure

Let $\{E_\alpha\}$ be orthogonal projections (sectors), typically fixed in time.

4.1 Sector invariance (discrete)

If $[U_p(t), E_\alpha] = 0$ for all $p \le P$, t, α, and if additionally T respects sectors (e.g. $T(E_\alpha x) = E_\alpha T(x)$), then each sector is invariant:

$$ X_0\in E_\alpha H\ \Rightarrow\ X_t\in E_\alpha H\ \forall t. $$

4.2 Time-varying projectors (warning)

If $E_\alpha(t)$ varies with time, pointwise commutation $[U_p(t), E_\alpha(t)] = 0$ does **not** by itself imply invariance across time; a moving-frame connection term is generally required (continuous time) or the sector notion must be reframed.

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5. Quantum embeddings (physically correct)

5.1 Open-system / randomized control (CPTP, “sum is native”)

Let ρ be a density operator. Choose unitaries $U_p(t)$ and weights $w_p(t) \ge 0$ with $\sum_{p\le P} w_p(t) = 1$. Define

$$ \mathcal{E}_t(\rho)=\sum w_p(t)\,U_p(t)\rho U_p(t)^\dagger. $$

To inject multiplicity while preserving complete positivity and trace, blend convexly with another CPTP map $\mathcal{F}_t$:

$$ \rho_{t+1}=(1-\lambda(t))\,\mathcal{E}_t(\rho_t)+\lambda(t)\,\mathcal{F}_t(\rho_t), \qquad \lambda(t) \in[0,1]. $$

Interpret $\lambda(t)$ as the **multiplicity gain**. If $\Lambda_m(t) \in \mathbb{R}$ is unconstrained, map it to $[0,1]$ via a squashing function $\lambda = \sigma(\Lambda_m)$.

5.2 Closed-system control (unitary, “product is native”)

Let G_p be skew-adjoint generators (e.g. $-iH_p$). Over a short step h, define the prime-ordered split propagator

$$ U(\tau+h,\tau)\approx \prod_{p\le P}^{\uparrow}\exp\big(h\,u_p(\tau)\,G_p\big). $$

A higher-accuracy palindromic (Strang-like) prime ordering is

$$ \Big(\prod_{p\le P}^{\uparrow} e^{\frac{h}{2}u_pG_p}\Big)\Big(\prod_{p\le P}^{\downarrow} e^{\frac{h}{2}u_pG_p}\Big). $$

Noncommutation budget

Leading splitting errors are governed by commutators $[G_p, G_q]$. A practical runtime metric is

$$ \mathrm{CommCost}(\tau):=\sum_{p<q}|[u_p(\tau)G_p,\,u_q(\tau)G_q]|. $$

Multiplicity coupling in closed control: use Λ_m to modulate h or u_p (not to add non-unitary terms unless dissipation is intended and modeled via CPTP composition).

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6. Canonical choices for prime channels

To make “primes” a testable inductive bias (not a label), choose U_p from canonical prime-structured families, e.g.

- prime-power averaging / Reynolds operators Π_{p^r}

- prime-indexed projectors induced by cyclic subgroup actions
- exponentials of prime-labeled Hamiltonian components $e^{hu_pG_p}$

These choices typically yield boundedness and interpretable multiscale structure.

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7. Predictions / expected outcomes (falsifiable)

7.1 Discrete contraction regime

If $q < 1$, then exponential forgetting holds:

$$ |X_t-Y_t|\le q^t|X_0-Y_0|. $$

7.2 Noncommuting control schedulability

For fixed step size h, fidelity degradation correlates with $\mathrm{CommCost}$. Palindromic prime order should systematically reduce error vs plain prime order when commutators are non-negligible.

7.3 Sector alignment tradeoff

Enforcing commutation with projectors yields strict sector invariance; relaxing it introduces controlled leakage measurable via drift/coherence metrics.

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8. Fastest validation path (three experiments)

Experiment A: Ledger-predicted convergence rate (nonlinear)

- $H = \mathbb{R}^n$ or $\mathbb{C}^n$, small prime set $\{2, 3, 5, 7\}$.
- Choose U_p, w_p, T (e.g. elementwise tanh).
- Sweep $\sup_t |\Lambda_m(t)|$ to move q across 1.
- Test: convergence/forgetting rate matches q.

Experiment B: Single-qubit noncommuting control

- $G_2 = -i\sigma_x$, $G_3 = -i\sigma_y$, $G_5 = -i\sigma_z$.
- Compare prime-ordered, palindromic prime-ordered, and baseline discretization.
- Test: fidelity loss correlates with $\mathrm{CommCost}$ and improves under palindromic ordering.

Experiment C: Canonical prime-channel advantage

- Use canonical prime-power projectors/averagers as U_p.
- Compare to Fourier/random baselines at equal parameter count.
- Test: improved multiscale separation metrics (sector purity, leakage control, reconstruction error).

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9. Reference implementation pattern (ledger enforcement)

Operationally, enforce the discrete-time ledger by clamping or rescaling to maintain $\|\Xi_P(t)\| + |\Lambda_m(t)| L_T \leq q_{\text{target}} < 1$ at runtime, while logging drift metrics.

(Implementation details intentionally omitted here; see accompanying code notes if needed.)

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